phosphate, is not converted to glucose 1 phosphate and accumulates in tissues and blood. The toxic galactose levels lead to enlarged liver, kidney failure and cataract.

The pathway of galactose metabolism is shown in figure 7.1.

Three types of galactosemia are known which are caused by mutations in GALT, GALK1 (galactokinase) and GALE (UDP-galactose 4′-epimerase) genes. However, GALT deficiency is the most severe and life threatening form of the disease (Figure.7.2).

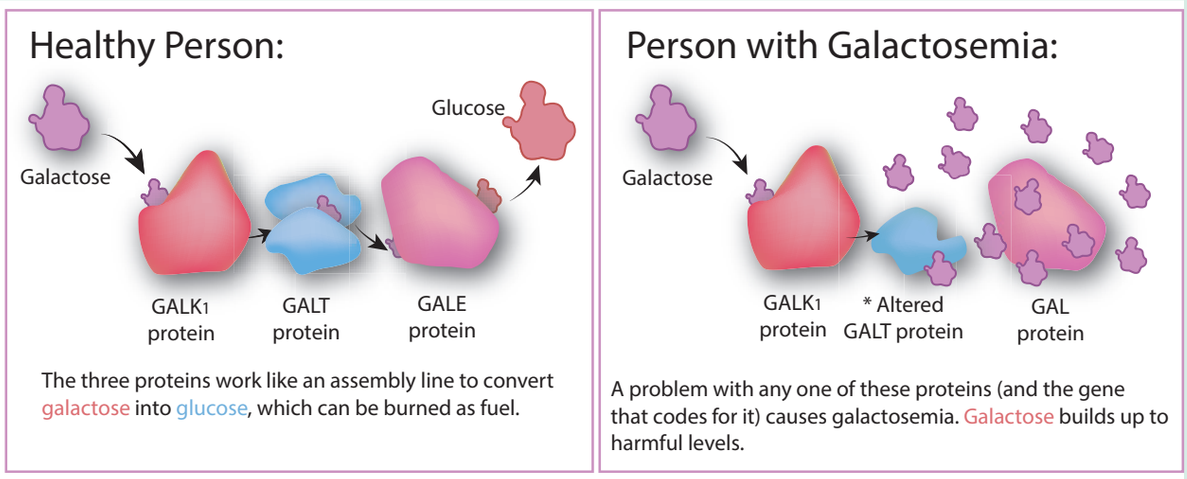


Figure.7.2 HealthyVs Galactosemia subjects

7.1.2 Symptoms

The deficiency of GALT is clinically important. Due to the enzyme defect, galactose accumulates in blood and is reduced by aldose reductase in the eye to the corresponding galactitol which causes cataract. The general condition is more severe, since, due to a deficiency in GALT, galactose 1- phosphate accumulates and injures the liver leading to its failure. Mental deterioration is caused due to accumulation of direct bilurubin, as it moves from blood to brain .

If an infant is breast feeding, but still experiences poor weight gain, it may due to galactosemia. Infants appear normal at birth but later they show failure to thrive and become lethargic. They have frequent vomiting and hypoglycemia. After 2 - 3 months of age, the liver may show fatty infiltration and lead to cirrhosis (nonfunctioning of liver cells). Galactosemia at this age is associated with mental retardation due to accumulation of galactose and galactose 1 - phosphate in cerebral cortex. Other symptoms include, jaundice, vomiting, hepatomegaly, *E.coli* sepsis and irritability.

7.2 VON GIERKE DISEASE

Glycogen storage diseases are a group of inherited disorders associated with glycogen metabolism. Von Gierke disease, also known as glycogen storage disease(GSD) type I, is one of the groups of rare genetic disorders due to the defect in one or more of the enzymes involved in glycogen metabolism leading to excessive accumulation of glycogen in the

XII U7 Inborn.indd 170 4/8/2020 3:22:38 PM